# 1 Estimation engine: the Kalman filter and RTS smoother

This section documents the recursive estimator at the core of the engine: a variable-dimension extended Kalman filter (EKF) with an optional prior-anchored iterated (IEKF) relinearisation pass, robust post-fit residual gating, and a backward Rauch–Tung–Striebel (RTS) smoother for post-processing modes. The state vector $\boldsymbol{x}$ is the uncombined PPP / PPP-RTK parameterisation of the differencing-transformations section of the Foundations document: receiver position (and optionally velocity), receiver clock and per-system time offsets, zenith wet troposphere and gradients, per-satellite slant ionosphere, receiver/satellite code and phase biases, and one float ambiguity per tracked carrier. Its dimension changes epoch to epoch as satellites rise and set and as ambiguities are reset; the layout and the dynamic key→index maps are owned exclusively by `filter/StateManager`, and the filter operates on that state without any fixed-index assumptions (`src/core/filter/KalmanFilter.cpp`).

Throughout we write $\boldsymbol{F}$ for the state-transition matrix, $\boldsymbol{Q}$ for the discrete process-noise covariance, $\boldsymbol{R}$ for the measurement-noise covariance, $\boldsymbol{S}$ for the innovation covariance, $\boldsymbol{K}$ for the Kalman gain, and $\boldsymbol{\nu}$ for the innovation (measurement minus predicted measurement). Predicted (a-priori) and updated (a-posteriori) quantities use the tilde/hat and $k|k-1$ / $k|k$ conditioning convention fixed in the notation-and-conventions section of the Foundations document.

## 1.1 Time update

**State transition.** The transition matrix $\boldsymbol{F}(\Delta t)$ for epoch interval $\Delta t = t_k - t_{k-1}$ is identity plus a small number of coupling blocks, assembled by `StateManager::getTransitionMatrix` (`src/core/filter/StateManager.cpp`). The only off-diagonal couplings are the kinematic integrators

$$\boldsymbol{F}[\,\boldsymbol{r},\boldsymbol{v}\,] = \Delta t\,\boldsymbol{I}_3, \qquad \boldsymbol{F}[\,c\mathrm{d}t,\ c\dot{\mathrm{d}}t\,] = \Delta t, \qquad \boldsymbol{F}[\,c\delta t^{(g)},\ c\dot{\delta t}^{(g)}\,] = \Delta t, \tag{1}$$

present only when the corresponding rate state exists (`hasVelocity()`, clock-rate, per-system rate). Every remaining state — ambiguities, per-satellite ionosphere, zenith wet delay, gradients, code and satellite biases, and any rate-free inter-system offset — has $\boldsymbol{F} = \boldsymbol{I}$ and is propagated as a pure random walk with its mean held constant. Under the BSSD (single-difference) parameterisation the clock and inter-system-offset rows are unobservable; their position rows in Equation (1) are suppressed and the corresponding cross-covariances are zeroed in place after propagation while the diagonal variance is preserved (`src/core/filter/KalmanFilter.cpp`).

The propagation itself is the standard EKF time update

$$\boxed{\begin{aligned}\tilde{\boldsymbol{x}}_{k|k-1} &= \boldsymbol{F}(\Delta t)\,\hat{\boldsymbol{x}}_{k-1|k-1}, &(2)\\ \tilde{\boldsymbol{P}}_{k|k-1} &= \boldsymbol{F}(\Delta t)\,\boldsymbol{P}_{k-1|k-1}\,\boldsymbol{F}(\Delta t)^{\mathrm{T}} + \boldsymbol{Q}(\Delta t), &(3)\end{aligned}}$$

implemented in `KalmanFilter::predict`. A non-positive $\Delta t$ is treated as a no-op (duplicate or out-of-order epoch), short-circuiting before Equation (3).

**Process-noise model.** The discrete process noise $\boldsymbol{Q}(\Delta t)$ is built from continuous spectral densities by `StateManager::getProcessNoiseMatrix`. Two block structures appear. Pure random-walk states have a scalar discrete variance increment

$$\boldsymbol{Q}_{ii} = q_i\,\Delta t, \tag{4}$$

where $q_i$ is the continuous variance rate ($m^2/s$) read from `filter.process_noise.*`. Representative entries are zenith wet troposphere (`troposphere_q`), the receiver code bias (`code_bias_q`), satellite phase bias (`sat_bias_q`), phase clock (`phase_clock_q`), and the float ambiguity (`ambiguity_q`). The ambiguity density defaults to a numerical safety floor ($10^{-12}$) rather than a physical value, and per-key overrides are supported for arc-specific reseeding. The per-satellite slant ionosphere uses a random walk whose zenith density is mapped to the slant by the squared obliquity factor,

$$\boldsymbol{Q}_{ii} = \frac{q_{\text{iono}}}{\sin^2\theta^i}\,\Delta t, \qquad \sin^2\theta^i \text{ floored at } 10^{-2}\ (\theta{\approx}5.7^\circ), \tag{5}$$

with $q_{\text{iono}}$ = `ionosphere_q` and optional per-satellite overrides driven by the adaptive-iono scheme.

Coupled position/velocity, clock/clock-rate, and inter-system offset/rate pairs use the continuous white-noise-acceleration (CWNA, integrated-random-walk) discretisation. For a coupled pair $(\xi, \dot{\xi})$ driven by acceleration PSD $q$,

$$\boldsymbol{Q}_{[\xi,\dot{\xi}]} = \begin{bmatrix} \frac{1}{3}\,q\,\Delta t^3 & \frac{1}{2}\,q\,\Delta t^2 \\ \frac{1}{2}\,q\,\Delta t^2 & q\,\Delta t \end{bmatrix}, \tag{6}$$

with $q$ given by `acceleration_psd_q` (position/velocity, the $3 \times 3$ block-diagonal extension of Equation (6) over the ECEF axes), `clock_rate_q`, and `system_time_rate_offset_q` respectively. When the velocity state is present the rate-free position density `position_q` is unused. Baseline (double-difference) mode reduces $\boldsymbol{Q}$ to the position block, a per-ambiguity random-walk term, and an optional differential-ZWD entry $\boldsymbol{Q}_{ii} = \sigma^2\Delta t$ governed by `baseline.differential_tropo_process_noise_m_per_sqrt_s`.

**Aided time update.** When inertial aiding is active, `KalmanFilter::predictImu` performs a strapdown mechanisation of the nominal position, velocity, and attitude and splices a $15 \times 15$ INS error-state transition $\boldsymbol{\Phi}$ and noise $\boldsymbol{Q}_{\text{ins}}$ — with error order $[\delta\boldsymbol{p},\ \delta\boldsymbol{v},\ \delta\boldsymbol{\psi},\ \delta\boldsymbol{b}_a,\ \delta\boldsymbol{b}_g]$ — into the GNSS-dimensioned $\boldsymbol{F}$ and $\boldsymbol{Q}$ over the integrated interval, after which Equation (3) runs on the full state with explicit symmetrisation $\boldsymbol{P} \leftarrow \frac{1}{2}(\boldsymbol{P} + \boldsymbol{P}^{\mathrm{T}})$. The generalised `predictAided` path applies the same splice for arbitrary aiding sources and delegates byte-identically to the GNSS-only Equation (3) when no aiding payload is present.

### 1.2 Measurement update

The measurement update (`KalmanFilter::update`) is a *batch* update: all observation rows of the epoch are processed simultaneously, not sequentially/scalar. A sparse-$\boldsymbol{H}$ optimisation accelerates the four design-matrix products ($\boldsymbol{HP}$, $\boldsymbol{HPH}^{\mathrm{T}}$, $\boldsymbol{KH}$, and the $\boldsymbol{H}$-touching parts of the gain), and is numerically identical to the dense computation to within a $10^{-12}$ structural threshold (`src/core/filter/KalmanFilter.cpp`).

Given the design matrix $\boldsymbol{H}$ from the linearisation of the code observation equation and phase observation equation of the Foundations document (position rows $-(\boldsymbol{e}^i_{\hat{r}})^{\mathrm{T}}$, the linearised geometric range there), the innovation $\boldsymbol{\nu}$ (`measurement.innovations`), and the measurement-noise covariance $\boldsymbol{R}$ — a full matrix when supplied, otherwise the diagonal $\boldsymbol{R} = \mathrm{diag}(\sigma^2_{r,i,s})$ of the stochastic noise model there — the innovation covariance and gain are

$$\boldsymbol{S} = \boldsymbol{H}\,\tilde{\boldsymbol{P}}_{k|k-1}\,\boldsymbol{H}^{\mathrm{T}} + \boldsymbol{R}, \tag{7}$$

$$\boldsymbol{K} = \tilde{\boldsymbol{P}}_{k|k-1}\,\boldsymbol{H}^{\mathrm{T}}\,\boldsymbol{S}^{-1}. \tag{8}$$

The gain is never formed by explicitly inverting $\boldsymbol{S}$. Exploiting the symmetry of $\boldsymbol{S}$, the engine factorises it once with an $\mathrm{LDL}^{\mathrm{T}}$ decomposition and recovers $\boldsymbol{K} = \left(\boldsymbol{S}^{-1}\boldsymbol{H}\tilde{\boldsymbol{P}}\right)^{\mathrm{T}}$ from a single solve

against $\boldsymbol{H}\tilde{\boldsymbol{P}}$ (`ldlt.solve(HP).transpose()`); on factorisation failure it falls back to a complete-orthogonal (pseudo-inverse) solve. The state update is

$$\delta\boldsymbol{x} = \boldsymbol{K}\,\boldsymbol{\nu}, \qquad \hat{\boldsymbol{x}}_{k|k} = \tilde{\boldsymbol{x}}_{k|k-1} + \delta\boldsymbol{x}. \tag{9}$$

**Joseph-form covariance update.** The covariance is updated in the numerically stabilised *Joseph* form, never the short form $(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\tilde{\boldsymbol{P}}$:

$$\boxed{\boldsymbol{P}_{k|k} = (\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\,\tilde{\boldsymbol{P}}_{k|k-1}\,(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})^{\mathrm{T}} + \boldsymbol{K}\,\boldsymbol{R}\,\boldsymbol{K}^{\mathrm{T}}.} \tag{10}$$

The Joseph form preserves symmetry and positive semi-definiteness for any gain, which matters because the engine deliberately departs from the optimal gain in its robust paths (Section 1.4); the result is symmetrised once more, $\boldsymbol{P} \leftarrow \frac{1}{2}(\boldsymbol{P} + \boldsymbol{P}^{\mathrm{T}})$, to clear round-off. The factor $(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})$ is assembled from the sparse $\boldsymbol{K}\boldsymbol{H}$ product with a unit shift on the diagonal. The same Joseph kernel, with the explicit gain $\boldsymbol{K} = \boldsymbol{P}\boldsymbol{H}^{\mathrm{T}}\boldsymbol{S}^{-1}$, is exposed as the pure static routine `josephPseudoUpdate` and reused for scalar pseudo-observation constraints (e.g. the GIM ionosphere prior via `applyPseudoObservation`) and for the IEKF pass of Section 1.3.

**Post-fit residuals.** After the update the post-fit residuals and their (diagonal) covariance are

$$\boldsymbol{r} = \boldsymbol{\nu} - \boldsymbol{H}\,\delta\boldsymbol{x}, \qquad S_{r,ii} = R_{ii} - \left(\boldsymbol{H}\,\boldsymbol{P}_{k|k}\,\boldsymbol{H}^{\mathrm{T}}\right)_{ii}, \tag{11}$$

of which only the diagonal of $S_r$ is materialised. The standardised residual that drives gating (Section 1.4) is

$$s_i = \frac{r_i}{\sqrt{\max\{\, S_{r,ii},\, R_{ii},\, 10^{-12} \,\}}}. \tag{12}$$

The denominator floor at the *working* $R_{ii}$ (the inflated noise actually used in $\boldsymbol{R}$, not the formal noise) is a deliberate robustness compromise: in ambiguity-absorption cases — double-differenced phase with an ambiguity gain near unity — the rigorous denominator $S_{r,ii}$ collapses toward $R\,\boldsymbol{S}^{-1}R$, which would register near-zero residuals as many-$\sigma$ outliers. The strictly correct standardisation uses $S_{r,ii}$ alone; the floor trades that rigour for stability and is documented as such in source.

### 1.3 Iterated relinearisation (IEKF)

For long initial position offsets a single linearisation of the geometric-range relation of the Foundations document per epoch can be insufficient. When `processing.relinearize_geometry` is enabled (and the run is not tightly-coupled IMU), the engine performs a prior-anchored iterated update — a Gauss–Newton / iterated EKF step — after the primary update.

**Anchor.** The iteration is anchored to the *stashed pre-update prior* $(\boldsymbol{x}^{-}, \boldsymbol{P}^{-})$ that the converged primary Equations (3) and (10) read, captured only when relinearisation is armed (`getLastUpdatePriorX` / `getLastUpdatePriorP`). This is the prior *after* any in-epoch ambiguity augmentation and outlier reset, so it is dimension-matched to the posterior; it is not a top-of-loop snapshot.

**Gauss–Newton step.** Let $\boldsymbol{x}_i$ be the current linearisation iterate (the committed state at which `measurement` was linearised, with innovations $\boldsymbol{z} - h(\boldsymbol{x}_i)$ and $\boldsymbol{H} = \partial h/\partial \boldsymbol{x}\big|_{\boldsymbol{x}_i}$). The prior-anchored iterate (`KalmanFilter::updateFromPrior`, via `josephPseudoUpdate`) is

$$\boxed{\begin{aligned} \boldsymbol{\nu} &= \big(\boldsymbol{z} - h(\boldsymbol{x}_i)\big) - \boldsymbol{H}(\boldsymbol{x}^- - \boldsymbol{x}_i), && (13)\\ \boldsymbol{K} &= \boldsymbol{P}^-\boldsymbol{H}^{\mathrm{T}}\big(\boldsymbol{H}\boldsymbol{P}^-\boldsymbol{H}^{\mathrm{T}} + \boldsymbol{R}\big)^{-1}, \qquad \boldsymbol{x}^+ = \boldsymbol{x}^- + \boldsymbol{K}\,\boldsymbol{\nu}, && (14)\\ \boldsymbol{P}^+ &= (\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\,\boldsymbol{P}^-\,(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})^{\mathrm{T}} + \boldsymbol{K}\,\boldsymbol{R}\,\boldsymbol{K}^{\mathrm{T}}. && (15) \end{aligned}}$$

The offset term $-\boldsymbol{H}(\boldsymbol{x}^- - \boldsymbol{x}_i)$ in Equation (13) re-expresses the innovation about the fixed prior and is what makes this a proper MAP / Gauss–Newton iteration: because $\boldsymbol{K}$ and $(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\boldsymbol{P}^-$ are always derived from the same fixed prior $(\boldsymbol{x}^-, \boldsymbol{P}^-)$, the covariance does *not* contract a second time and no information is double-counted.

**Firing condition.** The driver `Application::runGeometryRelinearization` (`src/core/app/Application.cpp` fires the iteration when all of the following hold:

1. `processing.relinearize_geometry` is on and a prior was stashed at the primary update;
2. the position change since the primary update exceeds a data-dependent threshold — `relinearize_dpos_threshol` (default 1 m) when phase is present, or `relinearize_dpos_threshold_code_only_m` (default 100 m) for code-only epochs;
3. the per-run cap `relinearize_max_count` ($-1$ = unlimited) is not exhausted.

The inner loop recomputes geometry and re-applies Equations (13) to (15) against the *fixed* converged exclusion set, for at most five iterations, declaring convergence when the position step falls below $10^{-4}$ m. Because GNSS geometry is nearly linear at the metre level, convergence is typically reached in a single pass and the threshold of item 2 means the iteration rarely fires at all. The tightly-coupled IMU configuration instead takes a single legacy *naive* second pass (re-applying the measurement against the posterior), since error-state attitude/bias feedback conflicts with a clean prior re-derivation.

## 1.4 Innovation and residual gating

The engine performs *no* $\chi^2$ innovation-covariance test. Outlier rejection (`KalmanFilter::validatePostFitResidua` operates on the standardised *post-fit* residuals $s_i$ of Equation (12) against a fixed scalar threshold, supplemented by robust median-based spatial detectors for pivot slips. The two top-level thresholds are compile-time constants, *not* JSON-tunable: the per-row rejection threshold is $s_{\max} = 6.0$ (`getResidualThreshold`) and the complete-reset trigger fraction is 0.5 (`getBadMeasPercent`); readers should not search for config keys for these.

**Per-row rejection.** If $\max_i |s_i| > 6.0$, the worst row is dispatched by type: a phase outlier triggers an ambiguity reset (with BSSD pivot handling and a sub-cycle $\lambda$-fraction floor), while a code outlier is excluded from the epoch. Correlated same-satellite phase rows on other bands are reset alongside it.

**Complete-reset backstop.** If the count of rejected rows exceeds $0.5\times$(original pre-peel observation count), the filter covariance is re-inflated (`resetFilterCovariance`) — a hard reset that, in the smoother, marks a segment boundary (Section 1.5).

**Robust pivot-slip detectors (BSSD).** Before the per-row gate, single-difference phase rows are grouped by (pivot, signal) and screened for a coherent pivot slip using median / median-absolute-deviation (MAD) statistics: a forced pivot handover fires when a group is large enough, a majority of its rows exceed threshold, the group median exceeds a metre-scale floor, and the MAD-to-median ratio is small (a tight cluster indicating a common-mode shift rather than scattered noise). Cross-band pooled and code-side variants apply analogous tests with their own floors, and non-pivot multi-band slips are confirmed by integer-cycle consistency ($|r|/\lambda_{\text{band}}$ rounding to the same nonzero integer across bands).

**Down-weighting (not gating).** Two paths inflate $\boldsymbol{R}$ rather than reject: a persistent EWMA-bias term adds the squared mean bias to $R_{ii}$ once an arc has warmed up, and a Huber post-fit scheme inflates $R$ for sub-cycle rows whose $|s_i|$ exceeds the Huber knee (innovation one-shot or post-fit IRLS variants, toggled by `enable_huber_r_inflation` and `enable_huber_innovation`). These modify the working $\boldsymbol{R}$ that feeds Equations (7) and (12), and are the reason the standardised residual floors at the working noise rather than the formal noise.

### 1.5 Backward RTS smoother

In post-processing modes, enabling `processing.run_smoother` runs a fixed-interval Rauch–Tung–Striebel recursion [1] over the stored forward trajectory (`include/filter/Smoother.hpp`, `src/core/filter/Smoothe`). The flag is force-disabled under `AppMode::RealTime`, since smoothing is non-causal and inherently a post-processing operation.

**Forward-pass capture.** Rather than reconstruct the a-priori from $\boldsymbol{F}$ and $\boldsymbol{Q}$ on the backward pass, the forward pass caches, per epoch, both the one-step prediction $\tilde{\boldsymbol{x}}_{k|k-1}, \tilde{\boldsymbol{P}}_{k|k-1}$ and the posterior $\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}$, together with $\boldsymbol{F}_k$ and the full set of core and dynamic key→index maps (`capturePriorSnapshot` / `capturePosteriorSnapshot` in `src/core/app/Application.cpp`). A capture-ordering invariant makes the stored prior exactly $\boldsymbol{F}_k\hat{\boldsymbol{x}}_{k-1|k-1}$: the a-priori is recorded immediately after `predict()` and before any cycle-slip reseeding or measurement update, and the a-posteriori is recorded *before* integer ambiguity resolution, so the smoother consumes *float* ambiguities only.

**Recursion.** Initialised at the terminal posterior $\hat{\boldsymbol{x}}_{N|N} = \hat{\boldsymbol{x}}_{N|N}$, $\boldsymbol{P}_{N|N} = \boldsymbol{P}_{N|N}$, the backward recursion is

$$\boxed{\begin{aligned} \boldsymbol{G}_k &= \boldsymbol{P}_{k|k}\,\boldsymbol{F}_{k+1}^{\mathrm{T}}\,\tilde{\boldsymbol{P}}_{k+1|k}^{-1}, && (16)\\ \hat{\boldsymbol{x}}_{k|N} &= \hat{\boldsymbol{x}}_{k|k} + \boldsymbol{G}_k\big(\hat{\boldsymbol{x}}_{k+1|N} - \tilde{\boldsymbol{x}}_{k+1|k}\big), && (17)\\ \boldsymbol{P}_{k|N} &= \boldsymbol{P}_{k|k} + \boldsymbol{G}_k\big(\boldsymbol{P}_{k+1|N} - \tilde{\boldsymbol{P}}_{k+1|k}\big)\boldsymbol{G}_k^{\mathrm{T}}. && (18) \end{aligned}}$$

The smoother gain $\boldsymbol{G}_k$ is realised as an $\mathrm{LDL}^{\mathrm{T}}$ solve — the predicted covariance is symmetrised, then $\boldsymbol{G}_k = \big(\tilde{\boldsymbol{P}}_{k+1|k}^{-1}(\boldsymbol{P}_{k|k}\boldsymbol{F}^{\mathrm{T}})^{\mathrm{T}}\big)^{\mathrm{T}}$ via `ldlt.solve(...).transpose()` — never an explicit inverse, and the smoothed covariance Equation (18) is symmetrised after the update. If the factorisation fails, the step is abandoned and epoch $k$ keeps its forward posterior.

**Variable-dimension reconciliation.** Because $\dim \boldsymbol{x}_k \neq \dim \boldsymbol{x}_{k+1}$ in general, Equations (16) to (18) are restricted at each step to the *common-key subspace* — the states present at both $k$ and

$k{+}1$, found by intersecting the dynamic key→index maps (`collectCommonIndices`). Let $\mathbf{\Pi}_k$ and $\mathbf{\Pi}_{k+1}$ select the common rows in each epoch's state space; the implemented state update is then

$$\hat{\boldsymbol{x}}_{k|N} = \hat{\boldsymbol{x}}_{k|k} + \mathbf{\Pi}_k^{\mathrm{T}}\, \boldsymbol{G}_{k,C}\big(\mathbf{\Pi}_{k+1}\hat{\boldsymbol{x}}_{k+1|N} - \mathbf{\Pi}_k\tilde{\boldsymbol{x}}_{k+1|k}\big), \tag{19}$$

with the analogous projection on Equation (18). States outside the subspace retain their forward a-posteriori value. A subtle indexing invariant: $\tilde{\boldsymbol{x}}_{k+1|k}$, $\tilde{\boldsymbol{P}}_{k+1|k}$ and $\boldsymbol{F}_{k+1}$ are sized for epoch $k$'s layout (the predict step does not grow the state), so they are addressed with epoch $k$'s common indices, whereas $\hat{\boldsymbol{x}}_{k+1|N}$, $\boldsymbol{P}_{k+1|N}$ live in epoch $k{+}1$'s layout.

**Segment boundaries at ambiguity resets.** Two mechanisms of differing severity terminate or trim backward information flow at ambiguity resets:

- A *complete* filter reset at $k{+}1$ (the covariance re-inflation of Section 1.4, recorded as `is_discontinuity`) is a hard segment boundary: it severs the dynamic link $\tilde{\boldsymbol{x}}_{k+1|k} = \boldsymbol{F}\hat{\boldsymbol{x}}_{k|k}$, so no $\boldsymbol{G}_k$ is meaningful and backward flow is terminated — epoch $k$ keeps its forward posterior.
- A *single-satellite* ambiguity reset is handled softly: that satellite's ambiguities are excluded from the common subspace for the $k \to k{+}1$ step, because the re-seeded large-variance prior at $k{+}1$ carries no information backward and smoothing it would only inject noise. The exclusion matches by satellite prefix and is intentionally slightly over-broad, always erring toward the safe forward fallback.

**Static-position degeneracy.** In static PPP the receiver position is a (near-)constant state, so the position-block smoother gain propagates essentially the same correction to every epoch and the smoothed position is degenerate — constant across epochs — *by construction*. This is expected, not a defect: the smoother's value accrues on the non-position states (troposphere, ionosphere, float ambiguities) and on kinematic runs where position carries genuine epoch-to-epoch dynamics.

## References


[1] H. E. Rauch, F. Tung, and C. T. Striebel. "Maximum likelihood estimates of linear dynamic systems". In: *AIAA Journal* 3.8 (1965), pp. 1445–1450. DOI: 10.2514/3.3166.